Free and Fair? A Zero-Cost Audit of Reliability and Bias in Free-Tier LLM Judges across Families and Tiers

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Abstract

Free-tier LLM APIs make large-language-model judges accessible to any budget, yet the reliability of these freely usable referees is largely uncharacterised. We audit ten open-weight judges served on a single free API tier, drawn from four model families (Qwen, DeepSeek, GLM, Kimi) plus two proprietary free-tier references under one protocol: 400 answer pairs spanning mathematics, factual QA, creative writing, and abstract explanation, each judged in both presentation orders, with a test–retest subset and ground-truth-controlled family-preference probes, followed by a pre-registered confirmatory battery, all at zero monetary cost. out. First, severity collapse: DeepSeek flash and flagship judges abstain with ties on 47–59% of ground-truth-clear pairs, dragging their accuracy as low as 0.32; a forced-choice prompt revision eliminates the abstentions entirely, so the collapse is prompt-contingent rather than model-intrinsic. Second, instruction insensitivity: in a four-arm prompt ablation (original rubric, no anti-verbosity sentence, softened, length-normalised) the longer-answer pick rate stays at 0.48 across all arms (interaction $p = 0.93$), and the pooled anti-length tilt does not survive question-level clustering, so the initial pass's "instruction backfire" is reclassified as a fragile descriptive pattern. Third, family-specific kinship: in the

main pass Qwen judges significantly avoid same-family answers (regression -0.64, $p = 2 \times 10^{-4}$) while GLM rewards them ($+1.10$, $p = 1 \times 10^{-4}$); on new, harder verifiable tasks the family-level effect does not replicate, with only a small-sample Kimi self-effect remaining (33% versus 2% baseline, $n = 12$, exploratory). Fourth, panels are decisiveness instruments, not debiasing ones: majority voting cuts abstention from 30% to 14–18% at flat decision quality while leaving the shared position-bias component unchanged, so order swapping remains necessary. We distill the results into six deployment guidelines and release the complete artifact—questions, prompts, 9,453 main-pass verdicts, 1,815 robustness-experiment verdicts, 560 verified answers on 140 new tasks, and analysis code—so the audit can be re-run as free-tier catalogues evolve.

Keywords: LLM-as-a-Judge, evaluation reliability, position bias, self-preference bias, free-tier models

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1. Introduction

Large language models (LLMs) are increasingly asked not only to produce answers but also to grade them. The LLM-as-a-Judge paradigm, in which a model compares or scores candidate responses in place of human annotators, has become a workhorse of modern language technology. It powers public leaderboards built on pairwise comparison [1], substitutes for human preference labels in instruction tuning pipelines, and provides cheap regression tests for retrieval-augmented systems, agents, and text generation services [2, 3]. As the scale of evaluation grows far beyond what human rating budgets can cover, the quality of machine judgement itself has become a first-order sci-

entific question.

The dominant practice rests on a small set of closed, frontier models such as GPT-4 and Claude. This reliance carries well-known costs: per-token fees that grow linearly with evaluation volume, restricted data governance, and reproducibility risks when a provider silently updates a checkpoint. Open models offer an appealing alternative. Their weights are public or freely hosted, and several providers expose capable open families through zero-cost API tiers, which makes them attractive to academic groups and small engineering teams. Whether these freely accessible models can actually serve as dependable judges, however, remains poorly characterized. Existing bias studies concentrate on closed judges from a handful of providers [1, 4, 5], and the largest recent audit of 21 judges covers nine commercial providers while leaving open-weight families at the periphery. Openness does not imply neutrality: a judge trained by the same organization that produced some of the candidates may share stylistic priors, tokenization habits, or even training data with those candidates, which creates a structural risk of family favoritism that earlier self-recognition studies only partially capture [6].

This paper presents a systematic anatomy of reliability and bias for freely accessible LLM judges. We assemble ten free-tier judges—open-weight checkpoints reached through a single zero-cost API tier—drawn from four model families (Qwen, DeepSeek, GLM, and Kimi), spanning multiple capability tiers within each family, and complement them with two proprietary free-tier judges (Agnes-2.5-Flash and Agnes-3.0-Flash) as a reference group, evaluating all of them under one unified protocol. Five generators, one per family plus a second Qwen tier, answer 200 questions from four domains: grade-

school mathematics with verifiable ground truth [7], adversarial factual QA with reference answers [8], open-ended creative writing, and abstract explanation. From these answers we construct 400 stratified answer pairs, and every judge scores every pair in both presentation orders, yielding 9,453 usable verdicts, with an additional test-retest subset of 640 repeated judgements on the open judges to probe stochastic stability.

The design isolates four bias families with matched measurements for every judge. Position bias is measured directly by comparing verdicts on swapped presentation orders, length bias by the conditional win rate of the longer answer, accuracy on ground-truth-clear pairs in the two verifiable domains, and family preference by a quality-controlled protocol that tracks how often a judge rewards a same-family answer when the reference answer proves that answer wrong. Inter-judge structure is summarized by pairwise Cohen's kappa [9], and all headline estimates carry question-level cluster bootstrap confidence intervals.

Four findings structure the results. *Severity collapse*: the DeepSeek flash and flagship judges return ties on 47–59% of ground-truth-clear pairs, silencing their discrimination and pushing ground-truth accuracy below chance, a failure mode invisible to compliance checks; a forced-choice prompt revision removes the abstentions entirely, so the collapse is prompt-contingent. *Instruction insensitivity*: across a four-arm prompt ablation the longer-answer pick rate moves only between 0.476 and 0.490, the original-versus-no-sentence interaction is noise ($p = 0.93$), and no per-judge length coefficient survives Holm correction, so the initial pass's anti-length tilt is reclassified as a fragile descriptive pattern rather than an instruction effect. *Lineage-specific kinship*:

in the main pass, Qwen judges avoid same-family answers (kin-win 37%, regression -0.64, $p = 2\times10^{-4}$) while GLM-5.3 rewards its own family's answers ($+1.10$, $p = 1 \times 10^{-4}$); on new, harder verifiable tasks (competition mathematics, unit-tested code, multiple choice) the family-level effect does not replicate, and only a small-sample Kimi self-effect persists (33% versus 2% baseline, $p = 0.006$, $n = 12$, exploratory). *Panels are decisiveness instruments*: majority voting cuts abstention from 30% to 14–18% at flat decision quality (0.834 versus 0.847) while passing the shared position-bias component through unchanged, so order swapping remains necessary.

Beyond measurement, we translate the findings into deployment guidance for practitioners who cannot afford commercial judge APIs. We quantify how majority panels of three or five open judges trade compute for accuracy and bias robustness, revisit the order-swapping protocol as a cheap debiasing device, and identify which free-tier models are safe to use as referees for which kinds of tasks. All prompts, question sets, raw verdicts, and analysis code are released with the paper to support reuse.

Our contributions are fourfold. First, we provide a reusable, zero-cost audit protocol—released question sets, probes, and screening thresholds—together with the first multi-family, multi-tier study of judging reliability for freely accessible LLM judges, covering four bias families under one protocol and contrasting open-weight judges with proprietary free-tier references. Second, we introduce a quality-controlled kin-preference measurement that separates family favoritism from genuine quality differences. Third, we give within-family tier-gradient evidence on order stability that links judge behaviour to model tier and serving configuration rather than to family iden-

tity alone. Fourth, we distill the results into concrete, near-zero-cost judging guidelines and release the complete experimental artifact so the audit can be re-run as free-tier catalogues evolve.

2. Related Work

2.1. LLM-as-a-Judge

Using LLMs to evaluate text quality was popularized by G-Eval, which framed NLG scoring as a probability-weighted form-filling task with chain-of-thought [2], and by MT-Bench, which established pairwise comparison with a reference-guided rubric as the default protocol and reported the first catalogue of judge failure modes [1]. Subsequent surveys systematized the design space of judges, prompts, and aggregation schemes [3]. A complementary line of work trains open models specifically for evaluation: the Prometheus family fine-tunes open LLMs on human feedback to produce reproducible graders whose agreement with humans rivals that of general-purpose judges [10]. Our study differs in orientation: rather than proposing a new judge or benchmark, we audit the judging behaviour of off-the-shelf open models that practitioners can already call for free, since in practice most teams use unmodified chat models as referees rather than dedicated evaluator checkpoints.

2.2. Judgement Biases

Three bias families dominate the literature. Position bias, the tendency to favour whichever answer appears first (or last), was demonstrated early for GPT-4-class judges and motivates order-swapping or consistency-based

mitigations [1, 11, 12]. Verbosity bias, the correlation between answer length and preference, was documented in RLHF preference labels and in arena-style evaluation, and motivated length-controlled win rates [13, 14]. Self-preference is the subtler tendency of a judge to favour its own generations: evaluators have been shown to recognize their own text and to reward it, even when the text carries a hidden watermark that the judge cannot consciously detect [6]. Comparative studies of human and machine judges further show that some biases, such as self-preference and conformity to anchor answers, are shared by humans and LLMs, while others, such as opinionatedness, differ sharply [4]. We generalize the self-preference lens from the individual model to the model family: because several providers ship many tiers of the same base model, a practical question is whether a judge favours any member of its own lineage, which we measure with ground-truth controls.

2.3. Reliability, Validity, and Judge Panels

A growing body of work asks whether judges agree at all. Cross-judge agreement with humans is often modest and task-dependent [15, 16]. A recent large-scale audit of 21 judges from nine providers reports that exact-match agreement overstates discriminative ability relative to Cohen's kappa, that judge rankings are unstable across benchmarks, and that high test-retest reliability can coexist with severe position bias [5]. Infrastructure work such as JudgeArena unifies judge benchmarks and ships tuned configurations for open judges [17]. On the mitigation side, panels of diverse judges have been proposed to average away idiosyncratic errors [18], though correlated errors within panels can undermine the diversity argument. Our contribution to this line is a family-and-tier-resolved view of agreement structure among free

judges, plus a panel analysis that respects family boundaries, so that panel gains are not confounded by kinship.

3. Experimental Setup

3.1. Model Roster

Table 1 lists the twelve judges and five generators. The ten open-weight judges are served through a single OpenAI-compatible endpoint (Alibaba Cloud Model Studio free tier) under identical API settings, which removes serving differences as a confound. The open roster spans four families: five Qwen checkpoints covering two flash generations, a 27B mid tier, a max tier, and a 2.4T-parameter Mixture-of-Experts flagship [19]; three DeepSeek checkpoints covering two flash tiers and a pro tier [20]; the GLM-5.3 flagship [21]; and Kimi-K3 [22]. The same five generators produce candidate answers, one flagship per family plus a second, deliberately smaller Qwen tier so that within-family quality variance exists for the kinship analysis. Two judges (Qwen3.8-2.4T and GLM-5.3) are served with mandatory reasoning, a platform restriction of the free tier; we account for this when interpreting their behaviour, and treat *thinking mode* as an observational variable rather than a confound to be hidden.

Beyond the open roster, we add two proprietary judges, Agnes-2.5-Flash and Agnes-3.0-Flash, accessed through their public API on the provider's zero-priced promotional tier. These models give a proprietary reference point against which the open judges can be read; they see the identical judging prompt, but their serving platform injects undisclosed system-level instructions ahead of it (observed as roughly 300–600 extra prompt tokens per call),

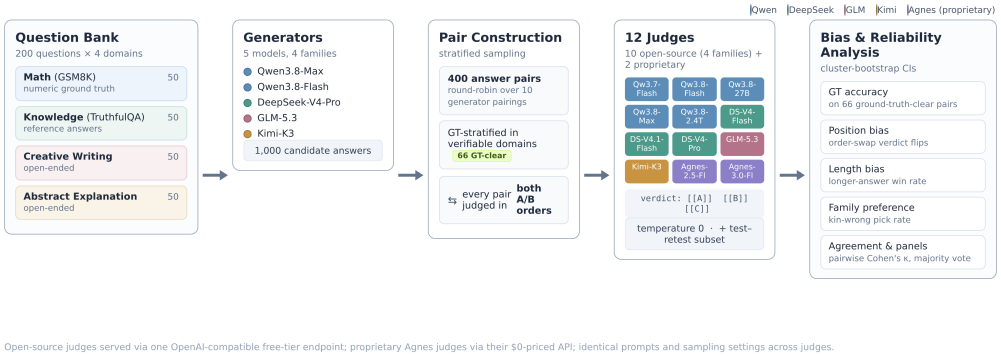


Figure 1: Overview of the evaluation pipeline. Two hundred questions from four domains are answered by five generators from four model families; stratified pair construction yields 400 answer pairs including 66 ground-truth-clear pairs, and twelve judges—ten open-source and two proprietary—evaluate every pair in both presentation orders, enabling direct measurement of position bias, length bias, family preference, accuracy, and inter-judge agreement.

so their absolute metrics are indicative rather than strictly comparable, a caveat we revisit in Section 7.

3.2. Task Domains and Questions

We evaluate on 200 questions, 50 per domain, chosen to combine verifiable and open-ended tasks. Mathematics uses grade-school word problems from the GSM8K test split, each with a certified numeric answer [7]. Knowledge uses TruthfulQA questions, which probe common misconceptions and carry reference answers [8]; this domain is deliberately adversarial so that generators exhibit meaningful quality differences. Creative writing consists of 50 fixed prompts covering stories, poems, letters, and other open-ended formats, authored for this study and released with the code. Abstract ex-

planation takes 50 recent computer-science abstracts retrieved from arXiv and asks for a plain-language, three-sentence explanation. The two verifiable domains enable objective accuracy measurement, while the two open-ended domains measure agreement where no ground truth exists, mirroring the conditions under which judges are most often deployed in practice.

3.3. Correctness Labelling

Answer correctness in the verifiable domains is determined in two stages. An initial pass labels at generation time: mathematics by extracting the final numeric answer with a regular expression (falling back to the last numeral in the text) and comparing it with the certified GSM8K answer under strict string equality, and knowledge by a tolerant keyword-containment heuristic against the TruthfulQA reference answer. A second pass refines both domains. For mathematics, the same extraction is retained but comparison switches to numeric equality after normalization, so that surface forms such as \$162, \$162.00, and 162 are treated as equal; this stage is fully deterministic and involves no language model, and it changed 8.4% of mathematics labels relative to the string-equality pass. For knowledge, each candidate answer is graded by a free-tier labelling model (Qwen3.7-Flash) that sees the question, the TruthfulQA reference answer, and the candidate answer, and returns a binary CORRECT/INCORRECT verdict with a one-sentence rationale at temperature zero; relative to the keyword heuristic this labeller changed 28.8% of knowledge labels, and manual spot checks confirmed the changes were almost always improvements.

We flag a circularity risk that this revision addresses directly: the knowledge labeller, Qwen3.7-Flash, is also a member of the judge panel, so the

knowledge labels that define ground-truth-clear pairs partly reflect the judgement of one model that is itself evaluated against them. Three mitigations apply. First, mathematics labels are fully deterministic and provide a conservative, circularity-free viewpoint: per-judge accuracy on mathematics ground-truth-clear pairs spans 0.11–0.60, notably below the pooled figures. Second, the revised protocol replaces the single labeller with two labellers from families outside the judge panel plus human adjudication of disagreements, reporting labeller agreement against the human gold standard. Third, all 66 ground-truth-clear pairs, including every label that changed between the two stages, are released in a human-verification worksheet with the full evidence trail.

3.4. Pair Construction

From the five generators' answers we form answer pairs by stratified sampling over the ten possible generator pairings. In the two verifiable domains we draw three pairs per question and preferentially pair a wrong answer against a correct one whenever the generators disagree under the correctness screen, which maximizes pairs with an objective quality difference; in the two open-ended domains we draw one pair per question in round-robin rotation. This produces 400 pairs in total, with every generator pairing occurring between 34 and 47 times. The presentation order of each pair is randomized once, and the judging protocol then evaluates every pair in both orders, which turns each pair into its own position-bias probe. In the verifiable domains, a pair is ground-truth-clear when exactly one side is labelled correct by the criteria of Section 3.3; 25% of mathematics pairs and 19% of knowledge pairs are ground-truth-clear (66 pairs in total); and only these pairs enter accuracy

and kin-preference analyses. Pair membership is fixed and released with the artifact, and all reported ground-truth quantities use the final labels.

3.5. Judging Protocol

Judges receive an MT-Bench-style rubric: an impartial-evaluator system message, the question, optionally the reference answer (for the verifiable domains only), and the two candidate answers labelled Assistant A and Assistant B. The prompt instructs the judge to weigh correctness, completeness, clarity, and instruction following, explicitly warns against rewarding length, and requires a one- or two-sentence justification followed by a machine-parseable verdict token: [[A]], [[B]], or [[C]] for a tie. Every judge evaluates every pair in both presentation orders at temperature zero, so any order-induced verdict change is attributable to position rather than sampling noise. Coverage is bounded by each model's free quota: two open judges exhausted theirs near the end of the main pass, and exact per-judge coverage is reported in Section 4. A stratified subset of 40 pairs (ten per domain) is judged twice more in the canonical order to measure test-retest stability. Verdicts are parsed by a strict bracket pattern with one reminder retry; the parse failure rate itself is reported as a compliance metric. To keep the judge input distribution uniform across domains, answers are capped at 900 characters and questions at 1,500 characters with an explicit truncation marker, a setting applied identically to every judge. The cap binds often: 76% of knowledge answers, 56% of creative answers, and 15% of mathematics answers exceed 900 characters, and for 11.6% of mathematics answers the extracted final numeric answer falls beyond the visible window; length-based statistics are therefore reported with both the full and the judge-visible length, and a

truncation-exclusion sensitivity analysis appears in the appendix.

3.6. Metrics

For each judge we compute the following quantities, all with 95% question-level cluster bootstrap confidence intervals over 2,000 resamples. Parse rate is the fraction of judgements that yield a valid verdict token. Tie rate is the fraction of canonical-order verdicts equal to [[C]]. Accuracy is computed on ground-truth-clear pairs as the fraction of verdicts that select the objectively correct side. Position bias contrasts the rate at which the first-presented answer wins among non-tie verdicts against the 0.5 null; the flip rate is the fraction of pairs whose canonically-mapped verdict differs between the two presentation orders. Length bias is the rate at which the longer answer wins among non-tie verdicts minus 0.5. Family preference is measured on kin pairs, where exactly one side comes from the judge's own family: the headline statistic is the kin-wrong pick rate, the fraction of ground-truth-clear kin pairs in which the judge selects its own family's answer although the reference answer proves it wrong. Inter-judge agreement uses pairwise Cohen's kappa on the three-class verdicts of canonical-order judgements [9]. A pooled logistic regression with judge fixed effects estimates the simultaneous effects of length difference, kinship, and an answer-quality difference indicator (derived from the correctness labels, so that both-correct and both-wrong pairs also inform the length and kinship coefficients) on the probability that Assistant A wins, providing a multivariate check on the univariate rates.

3.7. Zero-Cost Constraint

The entire study runs on free API quotas with hard spending protection enabled: every open-weight model is capped at its complimentary one-million-token allocation, after which the provider rejects further calls instead of billing them, and the two proprietary judges run on their provider's zero-priced promotional flash tier under platform rate limits. Token consumption was logged per model throughout, and the total footprint of the experiment, 8.4 million tokens across roughly 11,700 API calls, stayed within the complimentary budget. This constraint is not merely logistical; it demonstrates that a full multi-family judge audit is reproducible by any research group at zero monetary cost, which we view as part of the contribution.

4. Results

Two free-quota exhaustions shape the coverage picture and are reported transparently. GLM-5.3 exhausted its complimentary allocation after 714 of 800 main-pass judgements (89%; its coverage is complete for mathematics and nearly complete for knowledge, the two verifiable domains), and Qwen3.8-2.4T completed 782 of 800 (98%); both models could not participate in the test-retest subset, which therefore covers eight open judges. All remaining judges have complete coverage. Agreement statistics use pairwise-available pairs, so partial coverage degrades precision rather than introducing selection bias into the headline metrics. In total the study collected 9,453 usable main-pass verdicts and 640 test-retest verdicts. The proprietary judges completed the main pass in full, but their retest pass was curtailed by platform rate limits and is excluded from headline test-retest statistics; the raw

partial records are released with the artifact for transparency.

4.1. Compliance and Decision Severity

Instruction compliance is nearly universal: eleven of twelve judges parse successfully on at least 99% of calls, and the only notable exception is GLM-5.3 at 94.4%, whose mandatory reasoning phase occasionally consumes the reply budget before the verdict token is emitted; the one-reminder parse retry is invoked on only 1.2% of calls overall, rising to 9.4% for GLM-5.3. Compliance, however, is the weakest notion of reliability. Decision severity—how often a judge commits to a winner at all—separates the panel far more sharply, and it separates by family. The five Qwen judges tie on 15–31% of canonical-order pairs, whereas DeepSeek-V4-Flash and DeepSeek-V4-Pro tie on 50% and 59% respectively, GLM-5.3 on 47%, and Kimi-K3 on 44%. The two proprietary references bracket the open panel—Agnes-3.0-Flash ties on 28% of pairs, Agnes-2.5-Flash on 44%—so severity collapse is not an artifact of open-source models or of free tiers specifically; it is a serving-and-tuning choice that happens to differ systematically across providers. Because ties are excluded from conditional bias statistics and scored as errors in accuracy, this single parameter reshapes every downstream metric for the affected judges, as the following subsections show.

4.2. Ground-Truth Accuracy

Figure 2 reports accuracy on the 66 ground-truth-clear pairs. The Qwen family forms the top cluster (0.64–0.71), with the two flash checkpoints ahead of the larger tiers; GLM-5.3 reaches 0.61 and Kimi-K3 0.50. The DeepSeek family splits: DeepSeek-V4.1-Flash achieves a usable 0.56, but DeepSeek-

V4-Flash (0.42) and DeepSeek-V4-Pro (0.32) fall below the 0.5 chance line. DeepSeek-V4-Pro's below-chance score is a direct consequence of severity collapse: on 59% of pairs it abstains with a tie, which the strict scoring counts as an error. Its 0.32 is thus best read as "right one time in three, silent most of the rest" rather than as active misjudgement, and the same reading applies with less force to DeepSeek-V4-Flash. Within-family scale gradients are accordingly non-monotonic for DeepSeek (the flagship Pro tier is the *worst* family member) and roughly flat for Qwen, so capability tier alone is a poor predictor of judging usefulness—a finding we return to in Section 4.9.

4.3. Position Bias and Order Consistency

Four open judges show a statistically significant first-position preference (bootstrap CI excluding 0.5): Qwen3.8-27B at +0.113 and DeepSeek-V4.1-Flash at +0.146 are the strongest, with the two flash Qwens just past the significance line (+0.065 to +0.075). The remaining judges sit near zero, including the two largest Qwen tiers (−0.013 and −0.028). Average position bias is therefore mild by the standards of early GPT-4-class reports, but the order-swap flip rates expose substantial hidden instability: between 12.5% and 32% of pairs receive different canonical verdicts when the presentation order is swapped. Two judges served with mandatory reasoning are the most order-stable (Qwen3.8-2.4T at 0.125, GLM-5.3 at 0.138), and within the Qwen family the flip rate falls with tier (0.32 and 0.26 for the two flash generations, 0.20–0.31 for the mid and max tiers, 0.13 for the 2.4T flagship). Scale and thinking mode thus buy consistency rather than calibration.

4.4. Length Bias and the Anti-Verbosity Instruction

The judging rubric explicitly instructs models not to reward length, and in the naive pooled model of the main pass eight of ten open judges tilt against the longer answer (length bias -0.03 to -0.12; DeepSeek-V4-Pro at $+0.112$ is the only meaningfully length-favouring judge, and the proprietary references show the same signature, one of them in amplified form). That tilt is fragile, however: under question-level clustering the pooled coefficient is -0.46 per 1,000 characters of full length difference ($p = 0.022$) but -0.63 and non-significant ($p = 0.24$) for the length difference the judge actually sees, and none of the thirty per-judge length coefficients survives Holm correction. The robustness experiments (Section 5) settle the causal question directly: across a four-arm prompt ablation—original rubric, rubric with the anti-verbosity sentence deleted, a softened variant, and a length-normalised variant—the longer-answer pick rate stays between 0.476 and 0.490 and the original-versus-no-sentence interaction is noise ($+0.05$, one-sided $p = 0.93$). The anti-verbosity sentence neither causes nor corrects length behaviour on this benchmark; the apparent inversion of the verbosity bias should be read as a property of this roster and stimulus set, not of the instruction, and length behaviour must be probed directly with length-stratified pairs rather than managed by prompt editing.

4.5. Family Preference under Ground-Truth Control

Figure 4 and Table 2 summarise kinship behaviour, and the headline is heterogeneity: kinship effects are family-specific rather than universal. Pooling judges within family, Qwen judges pick a same-family answer in only 37% of non-tie kin pairs (CI $[0.31, 0.43]$, $n = 950$)—a significant kin-*avoidance*

(regression -0.64, $p = 2 \times 10^{-4}$)—while GLM-5.3 rewards its own family's answers ($+1.10$, $p = 1 \times 10^{-4}$); DeepSeek and Kimi show no family-level effect in the main pass (Fisher $p = 0.11$ and 0.16). The quality-controlled statistic sharpens the contrast for the large cells and stays directional for the small ones: when the reference answer proves the kin side wrong, Qwen judges still choose it in only 5.2% of cases ($n = 96$), GLM-5.3 in 27.3% ($n = 11$), and Kimi-K3 in 33.3% ($n = 9$); the GLM and Kimi kin-wrong samples are small, so we read them as directional evidence rather than precise estimates. GLM-5.3 is the sharpest illustration of preference overriding evidence: on the ground-truth-clear kin pairs where its own family is right it picks kin on every one, yet on those where its family is wrong it still picks kin more than one time in four, and the same pattern holds in the mathematics domain alone (kin-win 0.857, $n = 14$), which is free of the labeller circularity that affects the knowledge domain. These estimates pool family judges rather than individuals, and the pooled sign is a property of this roster's family mix, not a law of self-preference. The robustness experiments (Section 5) test whether these signals replicate on new, harder verifiable tasks.

4.6. Agreement Structure

Pairwise Cohen's kappa on canonical-order verdicts spans 0.27–0.72 with a median of 0.54 (Figure 5)—moderate agreement, broadly in line with published audits of commercial judges. Three structural patterns stand out. First, the highest-agreement pairs sit within the Qwen family: the two Qwen3.8 flash generations agree at $\kappa = 0.72$, and the max and 2.4T tiers at 0.70; the strongest cross-family pairs reach the same level (Qwen3.8-2.4T with GLM-5.3 at 0.70, GLM-5.3 with Kimi-K3 at 0.70). The open judges

agree with the proprietary references (mean $\kappa = 0.53$) slightly less than they agree with each other (0.54), so proprietary status is neither a guarantee of a distinct viewpoint nor a privileged one. Second, the family-blocked means reveal a cross-family fault line between Qwen and DeepSeek ($\kappa = 0.44$, the lowest open–open block) while GLM pairs strongly with Kimi (0.70) and with Agnes-2.5-Flash (0.64). Third, the judges whose severity collapsed are also the least agreeable ones: DeepSeek-V4-Pro's mean kappa of 0.43 and the minimum pair Qwen3.7-Flash versus DeepSeek-V4-Pro at 0.27 show that abstention behaviour, not disagreement about winners, is what most damages raw agreement statistics.

4.7. Test–Retest Stability

Repeating a 40-pair subset in canonical order at temperature zero yields agreement between repeats of 0.75–0.97 across the eight open judges with remaining quota; with $n = 40$ per judge these estimates carry wide intervals. DeepSeek-V4-Pro is the least self-consistent (0.75) and Qwen3.8-27B the most (0.97). For DeepSeek-V4-Pro the raw 0.75 mostly reflects the reproducibility of its abstentions: it is stable in returning ties, which inflates raw consistency while contributing nothing to discrimination, and its order-stability and decision quality are correspondingly weak. This decoupling of consistency from validity echoes recent multi-judge audits and cautions against using test–retest agreement alone as a certification of judge quality. The robustness experiments extend the measurement to a cross-session setting: a 60-pair retest conducted weeks after the main pass (Section 5) agrees with the original canonical verdicts at 0.85–0.92, comparable to the intra-session values, so temperature-zero stability persists across sessions for the

judges with remaining quota.

4.8. Panel Behaviour: Accuracy Gains without Debiasing

Figure 7 evaluates majority-vote panels of k judges by exact enumeration over all $\binom{10}{3} = 120$ and $\binom{10}{5} = 252$ combinations, scored under the unified convention (ties count as errors). A panel of three lifts ground-truth accuracy from a single-judge mean of 0.582 (best single 0.712) to 0.696 (CI $[0.533, 0.845]$)—level with, not above, the best single judge—and five judges add a further +0.029 to 0.725; panels drawn from all twelve judges including the proprietary references reach 0.693/0.721. The larger effect is on decisiveness: abstention falls from 30% (the single-judge mean tie rate on ground-truth-clear pairs) to 17.8% ($k = 3$) and 14.3% ($k = 5$), while decision quality on the pairs where the panel commits stays flat (0.834 single versus 0.846–0.847 panel). Panels are thus decisiveness instruments rather than accuracy miracles. Position bias, however, is neither amplified nor cancelled: on the same ground-truth-clear pair set, mean signed first-position preference is +0.037 for single judges, +0.036 for panels of three, and +0.040 for panels of five. Order-swap aggregation on top of the panel adds only +0.011 accuracy (0.707 versus 0.696) at higher abstention (0.194), so order swapping remains mandatory at the single-judge level while its marginal benefit above a panel is modest; panels should combine judges from different families and rely on aggregation.

4.9. Scale Trends within Families

Figure 6 plots accuracy, absolute position bias, and flip rate against within-family tier. The Qwen family shows a clean and, to our knowledge,

previously unreported gradient in *consistency*: flip rate falls from 0.26–0.32 across the flash and mid tiers to 0.13 for the 2.4T flagship, while accuracy stays within a narrow 0.64–0.71 band. The DeepSeek family inverts the intuitive ordering: the Pro flagship is the least accurate and least agreeable family member because of severity collapse. Family identity and tuning choices dominate raw parameter count as determinants of judge behaviour, which is why the practical guidance of the next section keys on screening rather than on scale.

5. Robustness Experiments

The revision adds a pre-registered confirmatory battery (analysis plan released with the artifact) that re-examines the headline claims under the same zero-cost constraint. All experiments reuse the main-pass prompt and truncation settings, and every call logs the model code, the platform-reported snapshot id, and the date. A snapshot-stability check first re-judged 20 stratified pairs from the main pass in both orders on two judges weeks after the original run: canonical-verdict agreement was 0.875 (Kimi-K3) and 0.925 (Qwen3.8-27B), both above the pre-registered 0.85 threshold, so main-pass and new data are pooled and the main-pass prompt arm serves as the ablation baseline.

Instruction ablation (E1).. Four prompt arms—the original rubric (P1), the rubric with the anti-verbosity sentence deleted (P0), a softened variant (P2), and a length-normalised variant (P3)—were run on 60 length-stratified pairs covering all four domains with four judges from three families (720 new verdicts; the P1 arm reuses the main-pass canonical-order verdicts). The result

is a clean negative: longer-answer pick rates are 0.481 (P1), 0.490 (P0), 0.476 (P2), and 0.480 (P3); the P1-versus-P0 length interaction is +0.05 (one-sided $p = 0.93$); and per-arm length coefficients span −0.03 to +0.09. The anti-verbosity sentence neither causes nor corrects length behaviour on this benchmark, and the main pass's anti-length tilt is accordingly reclassified as a fragile descriptive pattern (pooled clustering $p = 0.022$ full-length, $p = 0.24$ judge-visible; no per-judge coefficient survives Holm).

Tie ablation (E2).. Severity collapse was re-tested on 44 tie-prone pairs (selected before judging; by construction they contain no ground-truth-clear pair, which itself shows that ties concentrate on genuinely close answers) with four judges and two prompt revisions. Forcing a choice (T1: "a tie is not an available verdict") eliminates ties entirely for every judge (0.00 versus 0.36–0.86 under the original rubric; exact McNemar, Holm $p \leq 0.0002$). Tightening the definition of a tie (T2: ties only for substantive equivalence) removes the DeepSeek excess (0.50–0.61 versus 0.82–0.86, Holm $p \leq 0.003$) but leaves the Qwen judges unchanged (0.23–0.32, Holm $p \geq 0.22$). Severity collapse is therefore largely prompt-contingent, and the practical fix is a one-line prompt change rather than a different model.

Harder verifiable tasks and kin replication (E3/E4).. To test the kinship signals on harder, fully verifiable material, 140 new tasks were built from competition mathematics (MATH Levels 4–5 with numeric answers), unit-tested code completion (HumanEval), and hard-subject multiple choice (MMLU); correctness is decided by frozen deterministic parsers and executed unit tests, with no LLM labeller anywhere in the loop. Four generators answered all tasks (560 verified answers; in the second batch a second Qwen gen-

erator model code with its own free allocation replaced a quota-exhausted one, logged per call), and 93 ground-truth-clear pairs (50 Qwen-wrong, 31 DeepSeek-wrong, 12 Kimi-wrong) were judged by four judges spanning the three families with remaining quota. The main-pass family-level signals do not replicate at this difficulty: kin-side error rates are 9.1
16.7
7.7
self-favouritism either (Qwen self 6.7
single exception is Kimi, whose judges err on 4 of 12 kin-wrong pairs (33
signal we report as exploratory. The honest summary: family-level kin favouritism is weaker and less general than the main pass suggested; the main-pass evidence for GLM (+1.10) and Qwen (−0.64) stands as measured, but it does not transfer to harder verifiable tasks for the families we could re-test.

Deployment without references (E5).. Removing the reference answer from the judging prompt changes nothing measurable for the judges with with-reference twins (27 and 24 pairs respectively): paired accuracy differences are 0.00 for both (95
cluster-bootstrap CIs $[-0.19, 0.19]$ and $[-0.13, 0.13]$), with tie rates moving by at most +7 points. A third judge without a with-reference twin shows 77.5
the paired estimates. Free judges do not lean on the reference block for these verifiable domains.

Cross-session stability (E8).. A 60-pair retest weeks after the main pass, on three judges with remaining quota, agrees with the original canonical

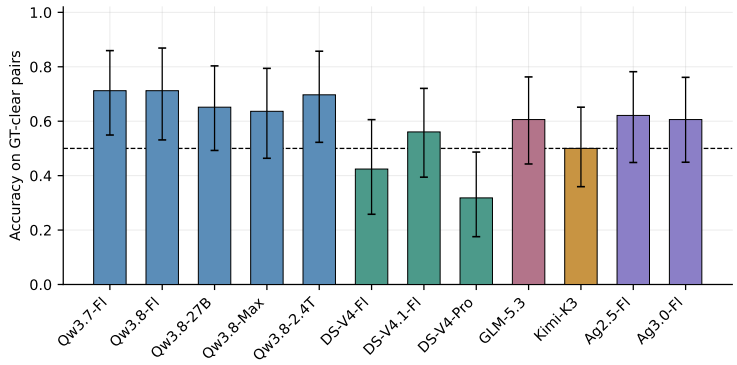


Figure 2: Ground-truth accuracy on the 66 ground-truth-clear pairs (canonical order), with question-level cluster bootstrap 95% CIs. The dashed line marks chance. The proprietary Agnes references are shown in purple.

verdicts at 0.85 (DeepSeek-V4-Flash), 0.87 (DeepSeek-V4.1-Flash), and 0.92 (Qwen3.8-27B)—level with the intra-session retest values despite the time gap.

Not run.. Two planned experiments could not be completed under the zero-cost constraint and are reported as such rather than claimed: a paid commercial-anchor run (E7, estimated at 2–5 US dollars for 800 calls) was deferred because the account held insufficient balance, and the thinking-mode contrast (E9) is unavailable because both forced-thinking models exhausted their free quota. A de-identified human-annotation kit (110 pairs plus 10 calibration pairs, three-way protocol, three annotators) is released with the artifact; its labels were still pending at revision time, so all reliability claims in this paper remain at the judge–judge level.

6. Discussion and Practical Guidelines

The audit yields one message above all: *free judges are usable, but only after screening, and never naively.* The properties that matter most are not the ones model cards advertise. Parameter count and flagship status predict neither accuracy nor severity; the DeepSeek Pro flagship is the least useful judge in the roster precisely because it abstains on most pairs, while a small flash checkpoint from the same provider discriminates better. What predicts behaviour is a combination of prompt contingency (severity), roster–stimulus properties (length), lineage (kinship, weakly), and deployment configuration (thinking mode for order stability). We distill the findings into six guidelines for zero-cost judging. We distill the findings into six guidelines for zero-cost judging.

G1: Screen for severity collapse first—and fix it with the prompt.. Before deploying any free judge, run a twenty-pair probe with a known quality difference and measure the tie rate. On ground-truth-clear pairs the main-pass tie rates are 47% (DeepSeek-V4-Flash) and 59% (DeepSeek-V4-Pro), with GLM-5.3 (32%) and Kimi-K3 (38%) near the line as threshold observations. A judge that abstains on clearly-different pairs will undermine any downstream use that assumes discrimination—leaderboards, win-rate estimation, best-of-n selection—regardless of how sound its individual verdicts are. The tie ablation shows that the cheapest remedy is a prompt revision, not a different model: forcing a choice eliminates ties entirely (Holm $p \leq 0.0002$), and tightening the tie definition removes the DeepSeek excess specifically.

G2: Prefer Qwen Max and, where quota allows, the 2.4T flagship as workhorses.. Within this roster, Qwen3.8-Max and Qwen3.8-2.4T offer the best combination of hard-flip stability (0.117 and 0.060; hard flips change the canonical verdict, soft ones only convert ties to decisions) and clean position behaviour. Two caveats: 2.4T is served with mandatory reasoning, emits at the output cap on 17.8% of calls, and consumed roughly double the tokens of the flash tiers, and both models' free-quota headroom is typically the scarcest in the Qwen family. Qwen3.8-27B, recommended in the initial pass, is demoted: under the two-way convention its first-position bias is Holm-significant at +0.139, the largest in the roster, so we no longer call it stable.

G3: Order swapping is mandatory for single judges; above a panel its extra benefit is modest.. Even the most order-stable judge flips on one pair in eight, and the least stable on one in three, so single-judge deployments must judge in both orders and aggregate on the canonical verdict. Above a majority panel, however, order-swap aggregation adds only +0.011 accuracy (0.707 versus 0.696) while raising abstention (0.178 to 0.194), so panels should combine judges from different families and balance positions across the pool rather than pay for per-pair double-judging of everything.

G4: Do not manage length behaviour by prompt editing.. The initial pass attributed an anti-length tilt to the anti-verbosity sentence; the four-arm ablation refutes the attribution: arm differences are within noise (interaction $p = 0.93$) and no per-judge length coefficient survives Holm correction. Prompt edits are the wrong lever for length behaviour on this benchmark. If length matters to your use case, probe it directly with length-stratified pairs and, where possible, standardise candidate lengths before comparison;

re-probe after any roster or task change, because the tilt appears to be a property of the roster–stimulus combination.

G5: Audit kinship with ground truth before trusting a judge on its own ecosystem.. Family preference is real but lineage-specific and, on harder verifiable tasks, weak: the main pass found Qwen avoidance (-0.64) and GLM reward ($+1.10$), while the replication on new tasks found no family-level effect for the re-testable families. Any evaluation that uses a judge from family X to rank models that include family X should run a ground-truth-controlled kin probe first. Size the probe for power: detecting a 30%-versus-8% kin-side error difference at 80% power needs about 50 kin-wrong pairs per family—roughly 0.1M free-tier tokens—not a token-sized spot check; our own Kimi cell ($n = 12$) illustrates how small cells produce exploratory-only results.

G6: Build panels across families, not across tiers.. The accuracy case for cross-family panels is weaker than the initial pass suggested—same-family, two-family, and three-family panels of three score 0.713, 0.697, and 0.688 with overlapping intervals and the ordering reversed—so we rest the recommendation on the agreement structure instead: family-level fault lines (Qwen–DeepSeek mean $\kappa = 0.44$) mean that a three-judge panel drawn from three families decorrelates errors far more effectively than three tiers of the same family, which matters because the shared position component passes through majority voting unchanged while the panel converts abstentions into decisions at zero monetary cost.

Relation to prior work.. Our findings refine three strands of the literature. The verbosity-bias literature documents that judges favour longer answers;

our four-arm ablation shows the standard mitigation sentence is behaviourally inert on this benchmark—neither causing nor correcting the tilt—and that the apparent inversion does not survive clustered inference, a caution for length-controlled win-rate claims that do not account for question-level clustering. The self-preference literature establishes that judges recognise and reward their own generations; we generalise the unit of analysis from model to family, show that the effect's *direction* varies by lineage, and add a replication on harder verifiable tasks in which the family-level effect does not transfer, which matters for the increasingly common situation where one provider both ships the judge and supplies some of the candidates. Finally, panel proposals argue that judge diversity averages away idiosyncratic errors; our panel analysis refines this: diversity cancels the independent noise component strongly enough to lift accuracy markedly, but the shared position component passes through majority voting unchanged, so panels must still pair diversity with order-balanced aggregation.

7. Limitations

Our findings should be read against several limitations. First, the ten open-source judges are all served through a single provider endpoint, and the two proprietary judges through a second one. Although this removes serving differences as a confound within each group, it also means that our observations reflect the providers' quantization and serving configurations of each checkpoint rather than the raw open weights, and replication on self-hosted inference may differ in detail; in particular, the Agnes platform injects undisclosed system instructions ahead of our judging prompt, so the

proprietary judges' absolute numbers are indicative rather than strictly comparable to the open panel. Second, the roster covers four model families that all originate from Chinese development teams, because these are the families with usable free tiers on the chosen platform at the time of the study; Western open families such as Llama or Mistral were not available under the zero-cost constraint, so our family-preference findings should be generalized to those families with care, and the GLM family is represented by a single checkpoint, so GLM-specific conclusions rest on one observation. Third, two judges are served with a mandatory reasoning phase, which is entangled with their family identity and tier; we treat thinking mode as an observational variable and report where it plausibly matters, but a fully crossed thinking-by-family design is not realizable on this platform. Fourth, the evaluation uses 400 pairs over four English-language domains. While the stratified design gives every judge the same stimulus set and the bootstrap accounts for question-level clustering, the absolute number of ground-truth-clear kin pairs per family remains modest, and kin-preference estimates for individual judges carry correspondingly wide intervals. Fifth, our generators produce single answers per question at a moderate sampling temperature; verdicts could vary with additional samples, and we leave multi-sample generation to future work. Finally, the two open-ended domains lack ground truth, so for those domains our reliability claims are about agreement among judges rather than closeness to correctness. The confirmatory battery has coverage limits that we report rather than paper over: the commercial anchor (E7) and the thinking-mode contrast (E9) could not be run under the zero-cost constraint, the kin replication cells for DeepSeek ($n = 31$) and Kimi ($n = 12$)

are small and exploratory, GLM-5.3 had exhausted its free quota and could not be re-tested on the new tasks, and the human-annotation kit released with the artifact was still unlabeled at revision time, so all reliability claims remain at the judge–judge level.

8. Conclusion

We audited the judging behaviour of every LLM reachable at zero monetary cost on two public platforms at the time of study: ten open-weight free-tier judges from four families and two proprietary free-tier references, evaluated under one protocol over 400 answer pairs in four domains, with both presentation orders, a test–retest subset, and ground-truth-controlled family-preference probes, followed by a pre-registered confirmatory battery (prompt ablations, harder verifiable tasks, reference-free deployment, and cross-session retests). The picture that emerges is nuanced but actionable. Free judges are viable: the strongest open checkpoints reach 0.71 accuracy on verifiable pairs and inter-judge agreement comparable to published commercial audits. But viability is conditional on screening, because the failure modes are systematic rather than random: severity collapse that is prompt-contingent and fixable with a one-line prompt change, an apparent anti-length tilt that proves fragile under clustered inference and inert to prompt revision, lineage-specific kin favouritism in the main pass that does not transfer to harder verifiable tasks, and majority voting that inherits, rather than removes, the shared position bias while converting abstentions into decisions. Every one of these failure modes is detectable with probes that cost a few thousand free-tier tokens, and every probe ships with the released artifact,

so the audit and its screening kit can be re-run by any group as free-tier catalogues evolve. As evaluation infrastructure comes to rest on machine judgement, the reliability of the judges themselves—especially the ones that cost nothing—deserves the same systematic attention that has been lavished on the systems they grade.

Data and Code Availability

All prompts, the 200-question evaluation set, 140 additional verified tasks, the deterministic parsers, the de-identified human-annotation kit, raw verdict logs for the main pass and the robustness battery (9,453 + 1,815 verdicts), and analysis scripts are released atscripts are released at `github.com/43aquarius/LLM-as-a-Judge`

Acknowledgements

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Table 1: Model roster. Judges evaluate all 400 answer pairs in both orders; generators produce the candidate answers. Tier gives the capability level within the family (F = flash/entry, M = mid, P = flagship/prime). Thinking indicates whether the serving platform enables a reasoning phase that cannot be disabled. Agnes judges are proprietary free-tier references (see text).

Model	Family	Tier	Role	Thinking
Qwen3.7-Flash (0715)	Qwen	F	judge	no
Qwen3.8-Flash	Qwen	F	judge + generator	no
Qwen3.8-27B	Qwen	M	judge	no
Qwen3.8-Max (0902)	Qwen	P	judge + generator	no
Qwen3.8-2.4T-A95B	Qwen	P+	judge	yes
DeepSeek-V4-Flash (0731)	DeepSeek	F	judge	no
DeepSeek-V4.1-Flash	DeepSeek	F	judge	no
DeepSeek-V4-Pro (0813)	DeepSeek	P	judge + generator	no
GLM-5.3	GLM	P	judge + generator	yes
Kimi-K3	Kimi	P	judge + generator	no
Agnes-2.5-Flash	Agnes[p]	F	judge	managed
Agnes-3.0-Flash	Agnes[p]	F+	judge	managed

[p]Proprietary judges, served via their own API.

Table 2: Per-judge summary on the main pass. Tie is the rate of [[C]] verdicts in canonical order; Acc is accuracy on the 66 ground-truth-clear pairs with ties counted as errors (the artifact adds the ties-excluded and ties-as-half conventions); PB is position bias (first-win rate among non-ties minus 0.5; under the two-way convention the Holm-significant first-position leans are Qwen3.7-Flash +0.093, Qwen3.8-Flash +0.069, Qwen3.8-27B +0.139, and DeepSeek-V4.1-Flash +0.128, with Qwen3.8-2.4T at −0.037 showing last-position preference); Flip is the order-swap verdict-flip rate; LB is length bias (longer-answer win rate among non-ties minus 0.5; no per-judge LB value survives Holm correction, so the column is descriptive); κ is the mean pairwise Cohen's kappa against the other judges; Retest is intra-session test–retest agreement on the repeated subset ("–" where the model exhausted its free quota or platform rate limits curtailed the retest; cross-session values from the robustness battery appear in Section 5).

Judge	Tie	Acc	PB	Flip	LB	κ	Retest
Qwen3.7-Flash	0.15	0.71	0.075	0.32	−0.072	0.46	0.85
Qwen3.8-Flash	0.29	0.71	0.065	0.26	−0.082	0.60	0.85
Qwen3.8-27B	0.24	0.65	0.113	0.31	−0.083	0.56	0.97
Qwen3.8-Max	0.24	0.64	−0.013	0.20	−0.063	0.57	0.95
Qwen3.8-2.4T	0.31	0.70	−0.028	0.13	−0.099	0.59	–
DeepSeek-V4-Flash	0.50	0.42	0.043	0.29	0.048	0.47	0.90
DeepSeek-V4.1-Flash	0.44	0.56	0.146	0.28	−0.034	0.55	0.82
DeepSeek-V4-Pro	0.59	0.32	0.070	0.25	0.112	0.43	0.75
GLM-5.3	0.47	0.61	0.019	0.14	−0.124	0.60	–
Kimi-K3	0.44	0.50	0.042	0.25	−0.051	0.58	0.90
Agnes-2.5-Flash^{p}	0.44	0.62	−0.052	0.27	−0.150	0.52	–
Agnes-3.0-Flash^{p}	0.28	0.61	−0.030	0.33	−0.057	0.54	–

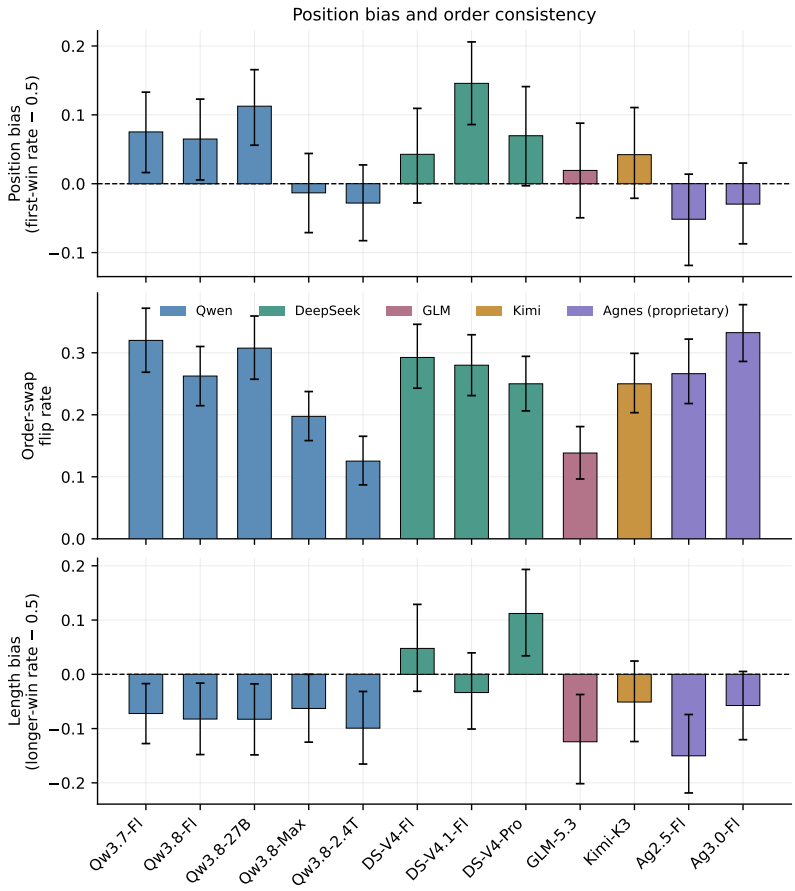


Figure 3: Bias trio per judge with 95% CIs: position bias (top), order-swap flip rate (middle), and length bias (bottom). Dashed lines mark the unbiased null.

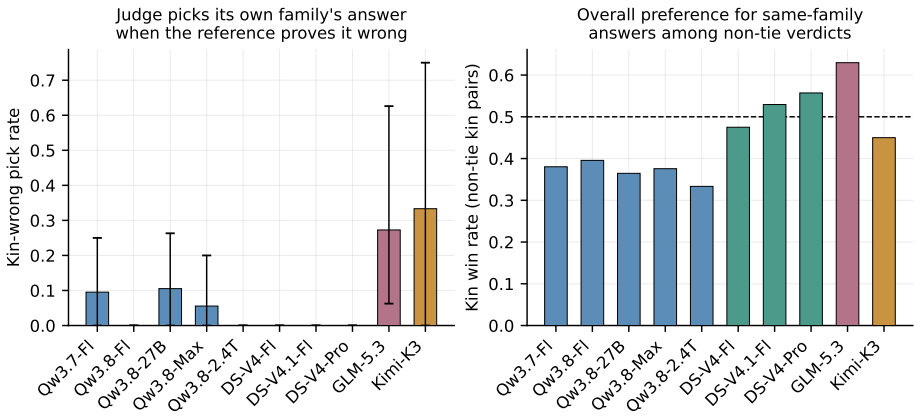


Figure 4: Family preference for the ten open judges. Left: kin-wrong pick rate, the fraction of ground-truth-clear kin pairs in which the judge picks its own family's answer although the reference proves it wrong. Right: overall kin win rate among non-tie kin pairs (dashed line: 0.5).

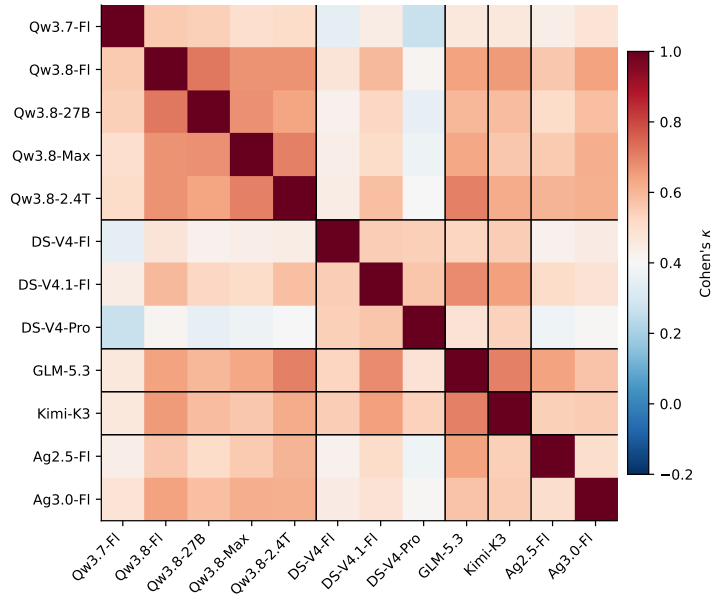


Figure 5: Pairwise Cohen's kappa between judges on canonical-order verdicts. Black grid lines separate model families.

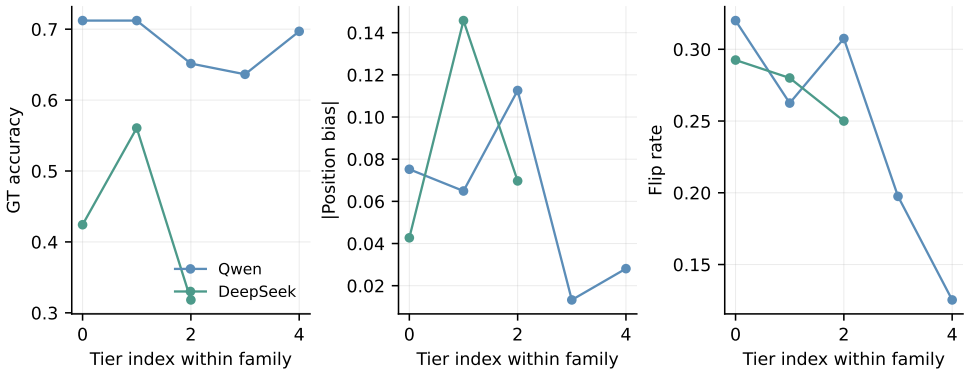


Figure 6: Within-family trends across capability tiers for families with at least two tiers: ground-truth accuracy (left), absolute position bias (middle), and order-swap flip rate (right).

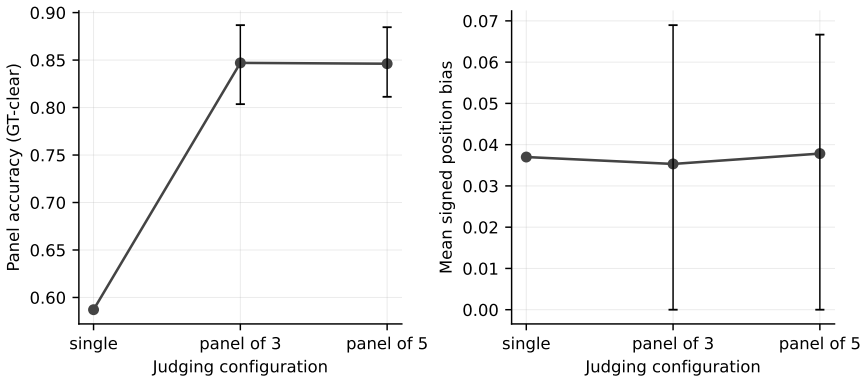


Figure 7: Majority-vote panels of open judges: ground-truth accuracy (left) and mean signed position bias (right) for single judges and panels of three and five. Panels raise accuracy but pass the shared position component through essentially unchanged.